

Self-Supervised Representation Learning for Multi-Class Diabetic Foot Ulcer Classification with Reliable Confidence Estimation

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ABSTRACT: The use of traditional supervised learning for DFU analyses is hindered by the size, heterogeneity, and imbalance of commonly available datasets. This study describes a framework for classification of DFU images using standard images and their associated clinical categories: none (not infected), infection, ischaemia, and both (all). It additionally explores the effectiveness of self-supervised representation learning (specifically, using a SimCLR model) to pre-train a model prior to subjecting it to supervised fine-tuning. The self-supervised model reached a macro-F1 score of about 0.923 compared to a macro-F1 score of approximately 0.881 for an ImageNet-initialised baseline. The performance of two semi-supervised approaches (FixMatch, pseudo-labeling) was less than the self-supervised method, suggesting both were negatively impacted by noise resulting from the pseudo-labels and the imbalance of the data. In this context, model calibration (ECE) and temperature scaling did not reduce accuracy but instead improved model probability reliability. Self-supervised pre-training can be a great alternative when there is limited amount of labelled data available, but naive semi-supervised learning can present challenges in the medical field.

Keywords: *Diabetic Foot Ulcer, Deep Learning, SimCLR, Self-Supervised Learning, EfficientNet, Medical Image Classification, Calibration.*

INTRODUCTION

The development of Diabetic Foot Ulcers (DFUs) is the most serious consequence of diabetes mellitus, affecting up to one in every fifteen diabetics during their lives and leading to most of the amputations performed on people with diabetes [1]. DFUs usually occur as a result of peripheral neuropathy, vascular insufficiency, repetitive trauma from the foot striking objects, and an impaired immune system. Many patients who develop DFUs lose their protective sensation gradually, and this process can go unnoticed until an infection or necrosis occurs. Patients who have a major amputation for DFU typically have a five-year mortality rate that ranges from 50% - 70% [1]. For this reason, it is critical to identify risks and assess DFUs using accurate methods as soon as possible, since this will ultimately reduce the number of amputations and improve healthcare outcomes for patients. The standard methods of documenting DFUs are still based on the visual observations made by a clinician using manual measurements and grading systems. The combination of how a clinician interprets a wound's appearance and their experience combined with variations in the type of lighting, quality of devices used to assess, variation in skin color, size (geometry) of ulcer, and the clinician's level of experience with DFUs creates significant inconsistency in how DFUs are ultimately diagnosed [12]. These inconsistencies become particularly evident in telemedicine environments and in health care systems that have high demand for DFU care; hence there is an urgent need for automated tools for DFU analysis that can facilitate objective, reproducible and standardized assessments that can be used to assist (not replace) clinicians' decision-making regarding patient treatment.

The advent of deep learning approaches to computer vision for this purpose has contributed significantly toward addressing this challenge. In particular, convolutional neural networks (CNNs) have proven very effective at detecting, classifying, and segmenting DFUs by automatically learning high-level representations directly from images [14]. Recently, there has been growing interest in applying Transformer-based vision models to DFU analysis since their attention mechanism allows for modeling long-range contextual relationships inherent in wound images, which can be challenging for conventional purely convolutional architectures [6],[15]. The emergence of publicly available DFU datasets (e.g., DFUC2020, DFUC2021 and DFUC2024) [5] has further accelerated progress on these models by providing standardized benchmark datasets for evaluation and comparison of new algorithms. Nevertheless, significant obstacles remain in this domain. Typically, DFU datasets will have only limited amounts of training samples due to the clinical expertise required for annotation as well as the time it takes to do so. The distribution of DFU cases is typically very unbalanced, with some DFU groups having very few cases relative to others. [7] Additionally, there are a lot of differences in imaging conditions at each clinical facility and the types of imaging devices that might be used to produce DFU images. These two factors can result in fully supervised models being overly fitted to the limited amount of labelled training data and cannot generalise effectively in practice after being deployed as many are not able to accurately reflect the diagnostic certainty. Additionally, clinical artificial intelligence systems also require accurate predictions as well as well calibrated probability estimates of their predictions to help inform their clinical decision making. In this research we will look to solve the challenge of 4 class DFU image classification (none, infected, ischaemia and both)

using the DFUC2021 dataset which is in contrast to many previous studies that have focused on binary or segmentation tasks and closely relates to the diagnosis categories clinically relevant. This study will also investigate representation learning for self-supervised feature learning using SimCLR in an effort to increase feature learning versus ImageNet initialised model training. Through self-supervised pre-training the network will use unlabelled DFU images to capture and learn domain specific visual structures prior to being trained with supervised methods.

We explore semi-supervised learning methods that utilize unlabelled data during training, however, our experiments show that they have a negative impact on performance when classes are imbalanced and when pseudo-labels are noisy. Finally, we tested temperature scaling for probability calibration in order to enhance the credibility of predicted confidence scores, while not affecting classification accuracy. Our results showed that self-supervised pre-training improved DFU classification performance compared to conventional supervised learning, and that calibrated confidence estimates improved their usability in clinical decision support systems.

RELATED WORK

Deep learning has rapidly advanced the automated assessment of diabetic foot ulcers (DFUs), particularly in the areas of detection, classification, and segmentation. Early DFU research focused primarily on convolutional neural networks (CNNs), which demonstrated strong capability in learning hierarchical visual features from clinical wound images. Yap et al [1] introduced one of the earliest CNN-based systems on the DFUC2020 dataset, but performance was limited due to challenges in detecting small or visually subtle ulcers. Cassidy et al [2] later applied hybrid preprocessing and CNN architectures to the same dataset, achieving improved accuracy but still requiring substantial manual annotation effort. Mahbod et al. [3] extended CNN based methods by combining U-Net segmentation with ResNet 50 classification to enhance DFUC2021 performance, even though this approach remained computationally intensive.

The release of the DFUC2024 dataset further stimulated research interest in segmentation-driven DFU modelling. Transformer-based architectures were introduced to better capture long-range spatial dependencies within wound regions. For example, Brodzicki et al. [8] demonstrated the effectiveness of Vision Transformer-based segmentation, although model complexity and computational cost remained limiting factors. Multi scale feature extraction methods have also been researched. Biswas et al. [11] employed multi-scale CNNs to capture wound-level and tissue-level structures, reporting strong Dice performance but sensitivity to edge ambiguity.

Self-supervised and semi-supervised learning have recently gained attention as strategies to mitigate the scarcity of labelled DFU data. Feature distillation-based representation learning for DFUC2024 was explored by Zhang et al. [9], while Wang et al.[10] reported semi-supervised segmentation performance improvements using SSL-enhanced U-Net. Ban et al.[4] further developed SSL-driven Mask R-CNN pipelines for ulcer segmentation. Although these works demonstrated promise, generalization across imaging environments remains an ongoing concern. Generative-adversarial-network (GAN)-enhanced segmentation approaches, such as those proposed by Kanti et al.[13], have achieved some of the highest Dice scores to date; however, GAN-instability and dependency on synthetic samples continue to pose risks for clinical deployment.

Beyond segmentation, several studies have emphasized clinically aligned DFU classification. Mahbod et al. [3] achieved approximately 70% accuracy on DFUC2021, while earlier DFU classification works reported accuracy in the 60–66% range 1–2. However, many studies focus on binary or two-class outcomes rather than multi-class disease stratification. Furthermore, few prior works evaluate probability calibration or uncertainty estimation, despite their importance in high-risk medical contexts.

METHODOLOGY

3.1 Overview of the Proposed Methodology

The objective of this project is to create a reliable and clinically significant deep learning model that can classify four different types of diabetic foot ulcers (DFUs), including no ulceration, infected ulcers, ischaemic ulcers and ulcers with both an infection and ischaemia. The methodology was designed to overcome several key issues in the analysis of DFUs (limited number of annotated images, unbalanced classes, and unreliable confidence estimates). Because of this, the methodology incorporates self-supervised learning to learn useful representations of DFU images, then fine-tune those representations using supervised learning to improve classification performance, and finally post-process the outputs from the model to produce calibrated probabilities.

The methodology involves four primary phases. First, we segment and organize DFU image data using stratified materials to maintain equal proportions of classes. Second, we train a baseline convolutional neural network (CNN) under supervised learning using pre-trained weights from ImageNet. Third, the model will have self-supervised pretraining with the SimCLR model to develop unique visual representations of DFUs using unlabelled images, then we will set the model's weights for

labelled data by using supervised fine-tuning. Finally, we will utilize temperature scaling to help calibrate the probability at which predictions occur so as to allow for more accurate predicted confidence scores. We also integrate semi-supervised learning methods, however, this approach was ultimately excluded from the final workflow due to their reduced overall performance. An overview of the entire work process is provided in Figure 1.

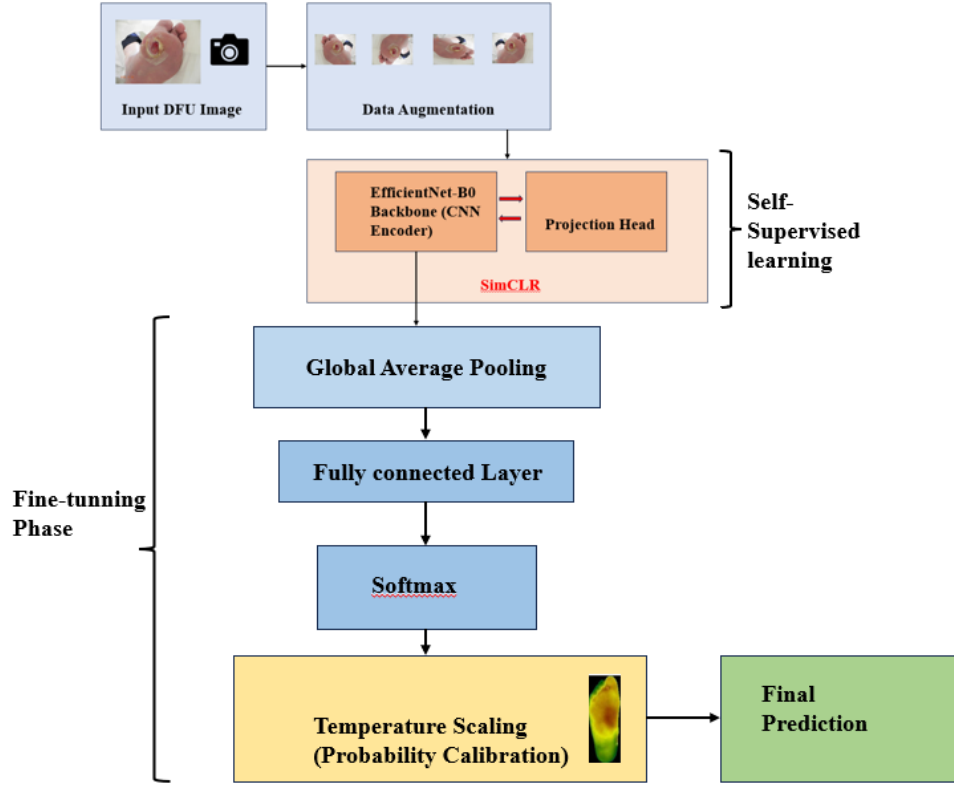


Figure 1: Overview of the proposed DFU classification methodology.

3.2 Dataset Description

Experiments are conducted by employing DFUC2021 dataset, which contains of RGB images of DFU's collected under clinical conditions. Images shows significant attributes in illumination, background clutter, size of ulcer, and skin color, reflecting realistic acquisition scenarios. Each image is annotated using a one-hot encoding scheme corresponding to four mutually exclusive clinical labels: *none*, *infection*, *ischaemia*, and *both*. These labels represent clinically meaningful diagnostic categories relevant to risk stratification and treatment planning. Images with missing, ambiguous, or inconsistent labels are excluded to ensure annotation reliability. The dataset exhibits strong class imbalance, with fewer samples in the ischaemia and combined-risk categories. To ensure fair evaluation, a stratified five-fold cross-validation strategy is employed, preserving the original class distribution across training and validation splits. Given the imbalance, macro-F1 score is selected as the primary evaluation metric, as it provides equal importance to all classes and better reflects minority-class performance compared with overall accuracy.

3.3 Network Architecture

The proposed classification network is built on a convolutional neural network backbone, EfficientNet B0, which was chosen due to its good trade-off between representational capacity and computational efficiency. EfficientNet B0 uses compound scaling to optimize network depth, width, and resolution jointly; thus, it can be used in real clinical deployments under resource constraints. The backbone network serves as a feature extractor and is followed by a global average pooling layer and a fully connected classification head that produces logits for the four DFU classes. In baseline supervised training, the backbone will be initialized with weights pretrained on ImageNet. For self-supervised learning, the classification head is removed and replaced by a projection head made up of fully connected layers as required by the SimCLR framework. After self-supervised pretraining, the projection head will be discarded while the pretrained encoder will be reused for supervised fine-tuning. This architectural design allows for efficient transfer of DFU-specific representations learned from unlabelled images to downstream classification tasks.

3.4 Self-Supervised Representation Learning

To minimize dependence on large annotated datasets, self-supervised representation learning is used based on the SimCLR framework. In this phase, the input image is augmented twice using stochastic transformations to produce two correlated views. These views are fed through a shared encoder and projection head, and a contrastive loss is applied to maximize the similarity between positive pairs while minimizing it with other samples in the batch. Self-supervised pretraining allows the model to capture invariant and discriminative features of DFU images without needing any manual annotations. After pretraining, the encoder weights are transferred to the supervised classification model which will allow this network to begin its supervised training from a DFU adapted representation space instead of generic ImageNet features.

3.5 Supervised Fine-Tuning and Semi-Supervised Evaluation

After the self-supervised pretraining, the model is fine-tuned with the labeled DFUC2021 images by adding a new classification head. Fine-tuning takes place with categorical cross-entropy loss and early stopping based on validation macro F1 to prevent overfitting. The performance is directly compared to the ImageNet initialized baseline to quantify how much self-supervised learning contributes. Also tested are some semi-supervised learning strategies like FixMatch and pseudo-labeling with exponential moving average (EMA) teachers but they always result in lower performances due to noise in pseudo-labels, amplification of class imbalances, and distortion of clinically relevant wound morphology. Therefore, semi-supervised learning has been left out from the final proposed framework and these results will be shared as negative but informative experimental findings.

3.6 Probability Calibration and Uncertainty Estimation

Reliable confidence estimation is crucial for clinical applicability. Hence, temperature scaling post hoc probability calibration is used, which rescales model logits without changing predicted class labels. The quality of calibration is determined through Expected Calibration Error (ECE) and reliability diagrams. Prediction entropy and confidence thresholds are also used to analyze uncertain cases that may need a clinician review. This calibration step enhances the interpretability and safety of the model when it is applied as a clinical decision support tool instead of an autonomous diagnostic system.

3.7 Implementation Details

The deep learning framework PyTorch is used to implement all experiments, and training is performed on NVIDIA GPUs. Input images are resized to 224×224 pixels and normalized with ImageNet statistics. The Adam optimizer is used for optimization, with fixed learning rate schedules across experiments to maintain an even playing field for comparison purposes. Checkpoints of the model are saved based on the validation macro F1 performance. Hyperparameters are chosen based on empirical methods and kept consistent between baseline and self-supervised experiments. The whole pipeline has been created in a manner that allows easy replication as well as computational efficiency.

RESULT

Table 1: Performance Comparison of Supervised, Self-Supervised, and Semi-Supervised Methods on DFUC2021

Method	Accuracy	Macro-F1
Baseline	0.8606	0.8809
SimCLR	0.9043	0.9229
FixMatch	0.6835	0.6570
FixMatch v2	0.7347	0.6244
Pseudo- labeling	0.7330	0.5960

The proposed diabetic foot ulcer classification framework was tested on the DFUC2021 dataset with a stratified validation split that preserved the distribution of its four clinically meaningful categories: none, infection, ischaemia, and both. Since there is an imbalance between classes in this dataset, macro F1 score will be used as the primary metric along with accuracy and per class performance and confusion matrices. Overall results for different learning strategies are summarized in Table 1. The supervised baseline model, EfficientNet B0 initialized with ImageNet weights, reached a best validation macro F1 of 0.8809 at an accuracy of 0.8606; hence it indicates strong performance but still shows some sensitivity to dataset imbalance. The class wise F1 scores produced by the baseline model were 0.8560 for none, 0.8492 for infection, 0.9130 for ischaemia, and 0.9055 for both with most confusion happening between none and infection classes. In order to better learn features under limited annotated data, the encoder was pretrained using SimCLR self-supervised learning followed by supervised fine tuning for four class DFU classification. This approach achieved the best overall performance with macro F1 of 0.9229 and accuracy of 0.9043 clearly outperforming the baseline. Class wise F1 improved across all categories to 0.8984 (none), 0.8948

(infection), 0.9474 (ischaemia), and 0.9512 (both). The confusion matrix of the SimCLR pretrained model is shown in Figure 2, highlighting strong diagonal dominance and reduced inter class confusion, including improved minority class recognition.

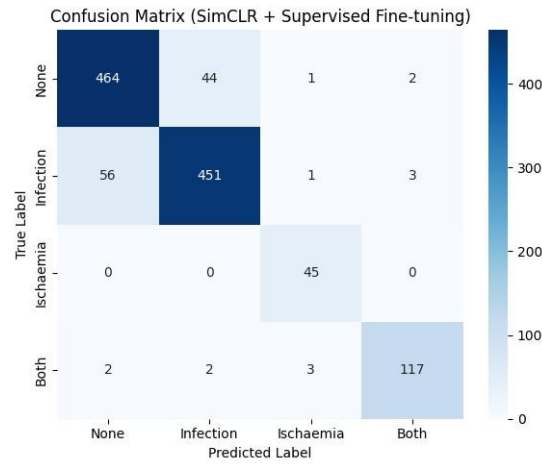


Figure 2. Confusion Matrix for SimCLR-Based DFU Classification Model

Semi supervised learning strategies were also tested but they constantly degraded performance. Fix-Match with EMA got a macro F1 of 0.6570 (accuracy 0.6835) and the stabilized Fix-Match variant achieved macro F1 of 0.6244 (accuracy 0.7347). Also, pseudo labeling with EMA gave a macro F1 of 0.5960 (accuracy 0.7330). These methods showed minority class collapse, which means pseudo label noise and imbalance amplification, so they were not included in the final framework (Table 1).

Finally, probability calibration was done with temperature scaling to better confidence reliability. Reliability diagrams before and after calibration are shown in Figure 3; this shows that scaling improves alignment between confidence and empirical accuracy. This calibration step did not change macro F1 or accuracy but improved interpretability of predicted probabilities, supporting uncertainty-aware clinical decision support. Generally speaking, the results indicate that SimCLR pretraining + supervised fine-tuning gives the best DFU classification performance while semi-supervised methods are unstable in this clinical dataset setting.

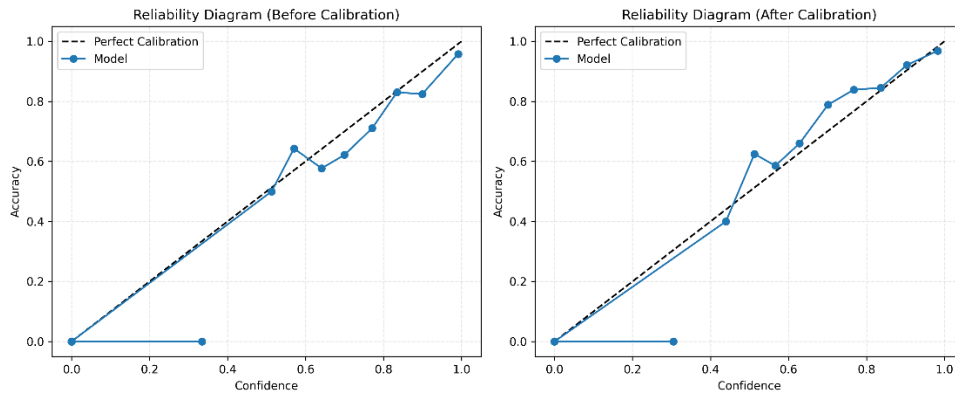


Figure 3. Reliability Diagrams Before and After Temperature Scaling for Model Calibration.

DISCUSSION:

The findings show that self-guided SimCLR pretraining greatly enhances DFU classification performance, with a macro F1 score of 0.9229 as opposed to the supervised baseline's 0.8809. All classes, including minority categories like ischaemia and both, show consistent improvement, suggesting improved feature representation and decreased interclass confusion. The model learns domain-specific visual patterns that are not captured by ImageNet pretraining alone by utilizing unlabeled data. Conversely, semi-supervised techniques like FixMatch and pseudo labeling reduced performance (macro F1 = 0.59–0.65), mostly because of class imbalance, pseudo label noise, and distortion of clinically significant characteristics. These results imply that without careful customization, naïve semi-supervised techniques are not suitable for DFU classification. In contrast to earlier research that reported 60–70% accuracy, Over 90% accuracy is attained by the suggested approach, which is a significant increase. This study covers multi-class DFU risk classification, which is more clinically significant than the majority of previous efforts that concentrate on segmentation or binary classification. Furthermore, temperature scaling

probability calibration increases confidence dependability, allowing for uncertainty-aware clinical decision assistance. All things considered, the suggested framework offers a reliable, effective, and practically useful method for DFU classification.

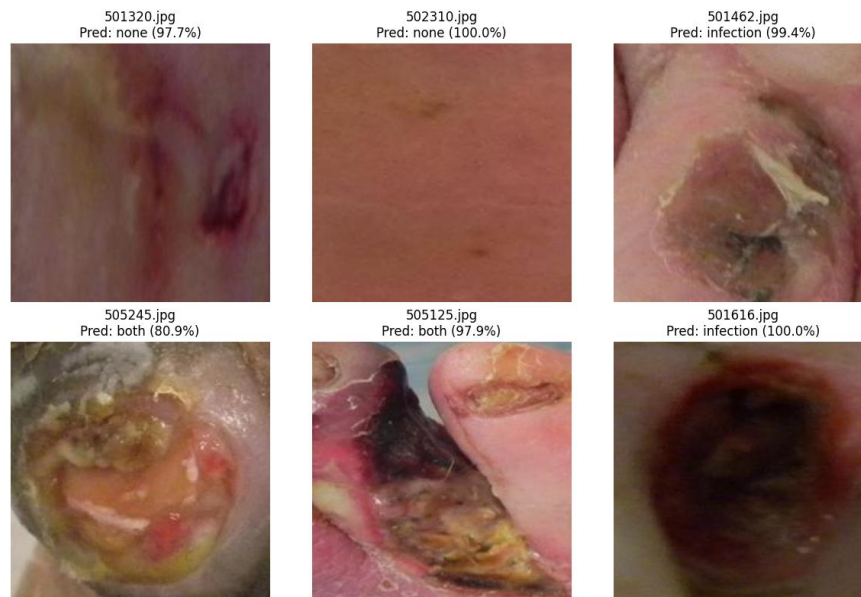


Figure 4. Qualitative Results Showing Model Predictions with Confidence Scores.

CONCLUSION:

This study used SimCLR-based self-supervised learning to propose a deep learning framework for multi-class diabetic foot ulcer (DFU) classification. The strategy successfully handled issues including class imbalance and a lack of tagged data. With a macro-F1 score of 0.9229, the SimCLR-pretrained model outperformed the supervised baseline and showed enhanced performance in all classes, including minority groups. On the other hand, pseudo-label noise and imbalance caused semi-supervised algorithms to perform worse. Furthermore, by calibrating confidence scores, temperature scaling increased clinical usefulness and prediction reliability. All things considered, the suggested framework offers a reliable, effective, and clinically applicable approach for DFU categorization, with great potential for practical implementation in decision-support systems.

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